Circle: Questions and Full Solutions

Question 01

English

Let $A = (-3, 0)$ and $B = (3, 0)$. Let R be the set of all points $P(x, y)$ with $y > 0$ for which $\angle APB \geq \dfrac{\pi}{3}$. Then the area of the region R is

(A) $8\pi - 3\sqrt{3}$
(B) $4\pi + 3\sqrt{3}$
(C) $8\pi + 3\sqrt{3}$
(D) $12\pi - 3\sqrt{3}$

Only one option is correct.

हिन्दी

माना कि $A = (-3, 0)$ तथा $B = (3, 0)$ हैं। माना R उन सभी बिन्दुओं $P(x, y)$ का समुच्चय है जिनके लिए $y > 0$ तथा $\angle APB \geq \dfrac{\pi}{3}$ है। तब क्षेत्र R का क्षेत्रफल है

(A) $8\pi - 3\sqrt{3}$
(B) $4\pi + 3\sqrt{3}$
(C) $8\pi + 3\sqrt{3}$
(D) $12\pi - 3\sqrt{3}$

केवल एक विकल्प सही है।

Solution

By the angle-in-a-segment theorem, all points on one side of AB that see AB under a fixed angle α lie on an arc of a circle through A and B.

A point inside that circle (on the same side) sees AB under a larger angle, and a point outside under a smaller angle. So for $y > 0$ the condition $\angle APB \geq \dfrac{\pi}{3}$ means that P lies inside or on the circle, above AB.

The chord $AB = 6$ subtends $2 \cdot \dfrac{\pi}{3} = \dfrac{2\pi}{3}$ at the centre O_1, so $AB = 2r \sin \dfrac{\pi}{3}$, giving $r = \dfrac{6}{\sqrt{3}} = 2\sqrt{3}$.

The centre lies on the perpendicular bisector $x = 0$ at distance $\sqrt{r^2 - 9} = \sqrt{3}$ from AB. For the arc above AB to see AB at an acute angle, the centre is on the same side, so $O_1 = (0, \sqrt{3})$.

Hence R is the part of the disc $x^2 + (y - \sqrt{3})^2 \leq 12$ with $y > 0$, that is, the major segment cut off by the chord AB.

The minor segment (below AB) has central angle $\dfrac{2\pi}{3}$ and area $\dfrac{r^2}{2}\left(\dfrac{2\pi}{3} - \sin\dfrac{2\pi}{3}\right) = 6\left(\dfrac{2\pi}{3} - \dfrac{\sqrt{3}}{2}\right) = 4\pi - 3\sqrt{3}$.

Area of the disc $= \pi r^2 = 12\pi$.

Area of $R = 12\pi - (4\pi - 3\sqrt{3}) = 8\pi + 3\sqrt{3}$.

Correct Option: (C)

Question 02

English

Consider all circles that pass through the point $(4, 0)$, cut off a chord of length 6 on the x-axis and cut off a chord of length 8 on the y-axis. Then the number of such circles for which the origin lies inside the circle is

(A) 2
(B) 4
(C) 1
(D) 0

Only one option is correct.

हिन्दी

उन सभी वृत्तों पर विचार कीजिए जो बिन्दु $(4, 0)$ से होकर जाते हैं, x-अक्ष पर लम्बाई 6 की जीवा तथा y-अक्ष पर लम्बाई 8 की जीवा काटते हैं। तब ऐसे वृत्तों की संख्या, जिनके अन्दर मूलबिन्दु स्थित है, है

(A) 2
(B) 4
(C) 1
(D) 0

केवल एक विकल्प सही है।

Solution

Write the circle as $x^2 + y^2 + 2gx + 2fy + c = 0$.

On the x-axis ($y = 0$): $x^2 + 2gx + c = 0$; the chord length is $|x_1 - x_2| = 2\sqrt{g^2 - c}$. So $2\sqrt{g^2 - c} = 6$, i.e. $g^2 - c = 9$.

On the y-axis ($x = 0$): $y^2 + 2fy + c = 0$; the chord length is $2\sqrt{f^2 - c} = 8$, i.e. $f^2 - c = 16$.

The circle passes through $(4, 0)$: $16 + 8g + c = 0$, so $c = -16 - 8g$.

Substituting in $g^2 - c = 9$: $g^2 + 8g + 16 = 9$, i.e. $g^2 + 8g + 7 = 0$, so $(g + 1)(g + 7) = 0$.

$g = -1$: $c = -8$ and $f^2 = 16 + c = 8$, so $f = \pm 2\sqrt{2}$: two circles.

$g = -7$: $c = 40$ and $f^2 = 56$, so $f = \pm 2\sqrt{14}$: two circles. In each case the radius squared $g^2 + f^2 - c$ is positive (1 and 65), so all four circles are real.

The origin lies inside a circle exactly when $S(0, 0) < 0$, and $S(0, 0) = c$.

For $g = -1$, $c = -8 < 0$: origin inside. For $g = -7$, $c = 40 > 0$: origin outside.

Hence exactly 2 circles contain the origin.

Correct Option: (A)

Question 03

English

For $\theta, \phi \in (-\pi, \pi)$ with $\theta \neq \phi$, let $P = (3\cos\theta, 3\sin\theta)$ and $Q = (3\cos\phi, 3\sin\phi)$ be points on the circle $x^2 + y^2 = 9$ such that $\tan\frac{\theta}{2}\tan\frac{\phi}{2} = \frac{1}{5}$. Then the locus of the midpoint of PQ is an arc of a circle whose radius is

(A) $\frac{3}{2}$

(B) $\frac{9}{4}$

(C) $\frac{9}{2}$

(D) $\frac{15}{4}$

Only one option is correct.

हिन्दी

$\theta, \phi \in (-\pi, \pi)$, $\theta \neq \phi$ के लिए माना कि $P = (3\cos\theta, 3\sin\theta)$ तथा $Q = (3\cos\phi, 3\sin\phi)$ वृत्त $x^2 + y^2 = 9$ पर ऐसे बिन्दु हैं कि $\tan\frac{\theta}{2}\tan\frac{\phi}{2} = \frac{1}{5}$ है। तब PQ के मध्यबिन्दु का बिन्दुपथ एक वृत्त का चाप है जिसकी त्रिज्या है

(A) $\frac{3}{2}$

(B) $\frac{9}{4}$

(C) $\frac{9}{2}$

(D) $\frac{15}{4}$

केवल एक विकल्प सही है।

Solution

The chord joining the points with parameters θ and ϕ on $x^2 + y^2 = 9$ is $x\cos\frac{\theta+\phi}{2} + y\sin\frac{\theta+\phi}{2} = 3\cos\frac{\theta-\phi}{2}$.

Put $y = 0$: the chord meets the x-axis at $x_0 = \frac{3\cos\frac{\theta-\phi}{2}}{\cos\frac{\theta+\phi}{2}}$.

Expand: $\cos\frac{\theta-\phi}{2} = \cos\frac{\theta}{2}\cos\frac{\phi}{2}\left(1 + \tan\frac{\theta}{2}\tan\frac{\phi}{2}\right)$ and $\cos\frac{\theta+\phi}{2} = \cos\frac{\theta}{2}\cos\frac{\phi}{2}\left(1 - \tan\frac{\theta}{2}\tan\frac{\phi}{2}\right)$.

Hence $x_0 = 3\cdot\frac{1+\frac{1}{5}}{1-\frac{1}{5}} = 3\cdot\frac{6}{4} = \frac{9}{2}$, the same for every admissible pair. So all chords PQ pass through the fixed point $A\left(\frac{9}{2}, 0\right)$, which lies outside the circle since $\frac{9}{2} > 3$.

Let M be the midpoint of PQ. The line from the centre O to the midpoint of a chord is perpendicular to the chord, so $\angle OMA = \frac{\pi}{2}$.

Therefore M lies on the circle with OA as diameter: $x\left(x - \frac{9}{2}\right) + y^2 = 0$, i.e. $x^2 + y^2 - \frac{9}{2}x = 0$.

Only the part of this circle inside $x^2 + y^2 = 9$ is traced (the chord must cut the circle), so the locus is an arc.

Its radius is $\frac{OA}{2} = \frac{9}{4}$.

Correct Option: (B)

Question 04

English

Let $T = (3, 4)$ and let L be the tangent to the circle $C_1 : x^2 + y^2 = 25$ at T. Let P be the point on L such that the length of the tangent drawn from P to the circle $C_2 : x^2 + y^2 - 4x + 2y + 1 = 0$ is equal to PT. Then PT is equal to

(A) $\sqrt{21}$
(B) $2\sqrt{5}$
(C) 4
(D) 5

Only one option is correct.

हिन्दी

माना कि $T = (3, 4)$ है तथा L वृत्त $C_1 : x^2 + y^2 = 25$ के बिन्दु T पर स्पर्श रेखा है। माना P रेखा L पर वह बिन्दु है जिससे वृत्त $C_2 : x^2 + y^2 - 4x + 2y + 1 = 0$ पर खींची गई स्पर्श रेखा की लम्बाई PT के बराबर है। तब PT बराबर है

(A) $\sqrt{21}$
(B) $2\sqrt{5}$
(C) 4
(D) 5

केवल एक विकल्प सही है।

Solution

The tangent to $x^2 + y^2 = 25$ at $T(3, 4)$ is $3x + 4y = 25$.

Any point P on this tangent sees C_1 along the tangent PT, so the length of the tangent from P to C_1 is PT, and $PT^2 = S_1(P) = x^2 + y^2 - 25$.

The length of the tangent from P to C_2 is $\sqrt{S_2(P)}$. The condition is $S_1(P) = S_2(P)$.

$S_1 - S_2 = (x^2 + y^2 - 25) - (x^2 + y^2 - 4x + 2y + 1) = 4x - 2y - 26$, so P lies on the radical axis $2x - y = 13$.

Solving $2x - y = 13$ and $3x + 4y = 25$: $y = 2x - 13$, $3x + 8x - 52 = 25$, $x = 7$, $y = 1$. So $P = (7, 1)$.

P is outside C_2: $S_2(7, 1) = 49 + 1 - 28 + 2 + 1 = 25 > 0$, so the tangent exists and has length 5.

$PT = \sqrt{4^2 + (-3)^2} = 5$.

Correct Option: (D)

Question 05

English

Let S be the focus of the parabola $y^2 = 4x$ and let C be the circle $(x-1)^2 + y^2 = 5$. The common tangents of the parabola and C touch the parabola at the points P and Q. Then the area of the triangle SPQ is

(A) 12
(B) 16
(C) 32
(D) 8

Only one option is correct.

हिन्दी

माना कि S परवलय $y^2 = 4x$ की नाभि है तथा C वृत्त $(x-1)^2 + y^2 = 5$ है। परवलय तथा C की उभयनिष्ठ स्पर्श रेखाएँ परवलय को बिन्दुओं P तथा Q पर स्पर्श करती हैं। तब त्रिभुज SPQ का क्षेत्रफल है

(A) 12
(B) 16
(C) 32
(D) 8

केवल एक विकल्प सही है।

Solution

The circle C has centre $(1, 0)$, which is the focus S of $y^2 = 4x$, and radius $\sqrt{5}$.

Every tangent of the parabola is the tangent at some point $(t^2, 2t)$: $ty = x + t^2$, i.e. $x - ty + t^2 = 0$.

Distance of $S(1, 0)$ from it: $\dfrac{|1 + t^2|}{\sqrt{1 + t^2}} = \sqrt{1 + t^2}$.

The line is tangent to C when this distance equals the radius: $\sqrt{1 + t^2} = \sqrt{5}$, so $t^2 = 4$, $t = \pm 2$.

So there are exactly two common tangents, $2y = x + 4$ and $-2y = x + 4$, touching the parabola at $P = (4, 4)$ and $Q = (4, -4)$.

Triangle SPQ: base $PQ = 8$ on the line $x = 4$, and the distance of $S(1, 0)$ from this line is 3.

Area $= \dfrac{1}{2} \cdot 8 \cdot 3 = 12$.

Correct Option: (A)

Question 06

English

Let $C_1 : x^2 + y^2 = 4$ and $C_2 : x^2 + y^2 - 12x + 26 = 0$. There are exactly two points P in the plane such that the polar of P with respect to C_1 coincides with the polar of P with respect to C_2, and exactly one of them lies outside C_1. Then the length of the chord of contact of the tangents drawn from this point to C_1 is

(A) $\sqrt{3}$
(B) $2\sqrt{3}$
(C) 3
(D) $2\sqrt{2}$

Only one option is correct.

हिन्दी

माना कि $C_1 : x^2 + y^2 = 4$ तथा $C_2 : x^2 + y^2 - 12x + 26 = 0$ हैं। समतल में ठीक दो बिन्दु P ऐसे हैं कि C_1 के सापेक्ष P की ध्रुवी, C_2 के सापेक्ष P की ध्रुवी के संपाती है, तथा इनमें से ठीक एक बिन्दु C_1 के बाहर स्थित है। तब इस बिन्दु से C_1 पर खींची गई स्पर्श रेखाओं की स्पर्श जीवा की लम्बाई है

(A) $\sqrt{3}$
(B) $2\sqrt{3}$
(C) 3
(D) $2\sqrt{2}$

केवल एक विकल्प सही है।

Solution

For $P(p, q)$ the polar with respect to C_1 is $px + qy = 4$.

With respect to C_2 ($g = -6$, $c = 26$) the polar is $px + qy - 6(x + p) + 26 = 0$, i.e. $(p - 6)x + qy + (26 - 6p) = 0$.

Two lines coincide when their coefficients are proportional: $\dfrac{p-6}{p} = \dfrac{q}{q} = \dfrac{26-6p}{-4}$.

If $q \neq 0$, the middle ratio is 1, so $p - 6 = p$, impossible. Hence $q = 0$, and the polars are the vertical lines $x = \dfrac{4}{p}$ and $x = \dfrac{6p - 26}{p - 6}$.

Equating: $4(p - 6) = p(6p - 26)$, so $6p^2 - 30p + 24 = 0$, i.e. $p^2 - 5p + 4 = 0$, giving $p = 1$ or $p = 4$.

So the two points are $(1, 0)$ and $(4, 0)$ (the limiting points of the two circles). $(1, 0)$ lies inside C_1 $(1 < 4)$, and $(4, 0)$ lies outside $(16 > 4)$.

For the external point $(4, 0)$ the polar with respect to C_1 is its chord of contact: $4x = 4$, i.e. $x = 1$.

The chord $x = 1$ of $x^2 + y^2 = 4$ has half-length $\sqrt{4 - 1} = \sqrt{3}$, so its length is $2\sqrt{3}$.

Correct Option: (B)

Question 07

English

For $\lambda \in \mathbb{R}$, let C_λ denote the circle $x^2+y^2-9+\lambda(x+y-1)=0$, and let $C : x^2+y^2-8x-4y+3=0$. For those values of λ for which C_λ and C intersect at two distinct points, let $\ell(\lambda)$ denote the length of their common chord. Then the minimum value of $\ell(\lambda)$ is

(A) $2\sqrt{17}$
(B) $2\sqrt{13}$
(C) 2
(D) 4

Only one option is correct.

हिन्दी

$\lambda \in \mathbb{R}$ के लिए माना कि C_λ वृत्त $x^2+y^2-9+\lambda(x+y-1)=0$ को निरूपित करता है तथा $C : x^2+y^2-8x-4y+3=0$ है। λ के उन मानों के लिए जिनके लिए C_λ तथा C दो भिन्न बिन्दुओं पर प्रतिच्छेद करते हैं, माना $\ell(\lambda)$ उनकी उभयनिष्ठ जीवा की लम्बाई है। तब $\ell(\lambda)$ का न्यूनतम मान है

(A) $2\sqrt{17}$
(B) $2\sqrt{13}$
(C) 2
(D) 4

केवल एक विकल्प सही है।

Solution

C_λ is the family of circles through the points where the circle $x^2+y^2=9$ meets the line $x+y=1$.

The common chord of two intersecting circles is their radical axis: $C_\lambda - C = 0$, i.e. $(8x+4y-12)+\lambda(x+y-1)=0$.

This is a family of lines through the intersection of $8x+4y-12=0$ and $x+y-1=0$. Solving $2x+y=3$, $x+y=1$: $K=(2,-1)$. (K is the radical centre of C, $x^2+y^2=9$ and any member C_λ.)

C has centre $O'=(4,2)$ and $R^2=16+4-3=17$. $O'K^2=(4-2)^2+(2+1)^2=13<17$, so K lies inside C; every line through K cuts C in a chord, and when the radical axis cuts C the two circles meet at those two points.

Among chords of C through K, the length $2\sqrt{R^2-d^2}$ is least when the distance d from O' is greatest, i.e. $d=O'K$, the chord perpendicular to $O'K$. Minimum length $=2\sqrt{17-13}=4$.

This chord is realised by some λ: the radical axis has normal $(8+\lambda,4+\lambda)$, which must be parallel to $\overrightarrow{KO'}=(2,3)$: $3(8+\lambda)=2(4+\lambda)$, $\lambda=-16$.

C_{-16}: $x^2+y^2-16x-16y+7=0$ has $r^2=64+64-7=121>0$, a real circle. Its radical axis with C is $2x+3y-1=0$, at distance $\dfrac{13}{\sqrt{13}}=\sqrt{13}$ from O', giving chord 4.

Hence the minimum value of $\ell(\lambda)$ is 4.

Correct Option: (D)

Question 08

English

Let $A = (-1, 3)$ and $B = (5, -5)$. Let S be the set of all real numbers k for which the line $2x - y = k$ contains a point P with $\angle APB = \frac{\pi}{2}$ but contains no point Q with $\angle AQB > \frac{\pi}{2}$. Then the product of all the elements of S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $A = (-1, 3)$ तथा $B = (5, -5)$ हैं। माना S उन सभी वास्तविक संख्याओं k का समुच्चय है जिनके लिए रेखा $2x - y = k$ पर एक बिन्दु P ऐसा है कि $\angle APB = \frac{\pi}{2}$ है, परन्तु रेखा पर कोई बिन्दु Q ऐसा नहीं है कि $\angle AQB > \frac{\pi}{2}$ हो। तब S के सभी अवयवों का गुणनफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Angle in a semicircle: $\angle APB = \frac{\pi}{2}$ exactly when P lies on the circle with AB as diameter and $P \neq A, B$.

A point Q inside this circle and not on the segment AB sees AB under an obtuse angle; a point outside sees it under an acute angle.

Diametric form: $(x + 1)(x - 5) + (y - 3)(y + 5) = 0$, i.e. $x^2 + y^2 - 4x + 2y - 20 = 0$, centre $(2, -1)$, radius $\sqrt{4 + 1 + 20} = 5$ (indeed $AB = \sqrt{36 + 64} = 10$).

If the line cut the circle at two points, the open chord between them would contain points Q inside the circle with $\angle AQB > \frac{\pi}{2}$ (only finitely many of them can lie on AB). So the line must be a tangent.

The tangent must touch at a point other than A and B. The tangents at A and B are perpendicular to AB (slope $-\frac{4}{3}$), so they have slope $\frac{3}{4}$, while $2x - y = k$ has slope 2. Hence every tangent of slope 2 touches at an admissible point P.

Tangency: distance of $(2, -1)$ from $2x - y - k = 0$ equals 5: $\frac{|5 - k|}{\sqrt{5}} = 5$, so $(k - 5)^2 = 125$, $k = 5 \pm 5\sqrt{5}$.

$S = \{5 + 5\sqrt{5},\ 5 - 5\sqrt{5}\}$ and the product is $25 - 125 = -100$.

Final Answer: -100

Question 09

English

For $\lambda \in \mathbb{R}$, let E_λ denote the curve $(x-1)(x-5)+(y-2)(y-6)+\lambda(x-y)^2=0$. There is exactly one value of λ for which E_λ represents a pair of straight lines, and for this value the two lines intersect at the point (α, β). Then the value of $\alpha - \beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\lambda \in \mathbb{R}$ के लिए माना कि E_λ वक्र $(x-1)(x-5)+(y-2)(y-6)+\lambda(x-y)^2=0$ को निरूपित करता है। λ का ठीक एक मान ऐसा है जिसके लिए E_λ सरल रेखाओं के एक युग्म को निरूपित करता है, तथा इस मान के लिए दोनों रेखाएँ बिन्दु (α, β) पर प्रतिच्छेद करती हैं। तब $\alpha - \beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$(x-1)(x-5)+(y-2)(y-6)=0$ is the circle with the segment from $(1,2)$ to $(5,6)$ as diameter; E_λ adds λ times the square of the line $x-y=0$.

Expanding: $x^2-6x+5+y^2-8y+12+\lambda(x^2-2xy+y^2)=0$, i.e. $(1+\lambda)x^2-2\lambda xy+(1+\lambda)y^2-6x-8y+17=0$.

Coefficients: $a=b=1+\lambda$, $h=-\lambda$, $g=-3$, $f=-4$, $c=17$.

A second-degree equation is a pair of straight lines when $\Delta = abc+2fgh-af^2-bg^2-ch^2=0$ (and $h^2-ab \geq 0$ for real lines).

$abc=17(1+\lambda)^2$, $2fgh=2(-4)(-3)(-\lambda)=-24\lambda$, $af^2+bg^2=25(1+\lambda)$, $ch^2=17\lambda^2$.

$\Delta = 17+34\lambda+17\lambda^2-24\lambda-25-25\lambda-17\lambda^2=-15\lambda-8$.

So $\Delta=0$ only for $\lambda=-\dfrac{8}{15}$. Then $h^2-ab=\dfrac{64}{225}-\dfrac{49}{225}=\dfrac{15}{225}>0$, so the lines are real and distinct.

For $\lambda=-\dfrac{8}{15}$, multiplying by 15: $7x^2+16xy+7y^2-90x-120y+255=0$.

The point of intersection makes both partial derivatives zero: $14x+16y-90=0$ and $16x+14y-120=0$. Solving: $x=11$, $y=-4$; it satisfies the equation $(847-704+112-990+480+255=0)$.

Check by geometry: the line $y=x$ cuts the circle (distance of centre $(3,4)$ is $\dfrac{1}{\sqrt{2}} < 2\sqrt{2}$), and the degenerate member is the pair of tangents at the two cutting points, meeting at the pole of $y=x$, which is $(11,-4)$.

$\alpha-\beta=11+4=15$.

Final Answer: 15

Question 10

English

Let C be the circle $x^2 + y^2 - 6x - 6y + 2 = 0$. Then the minimum value of xy over all points (x, y) on C is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि C वृत्त $x^2 + y^2 - 6x - 6y + 2 = 0$ है। तब C पर स्थित सभी बिन्दुओं (x, y) के लिए xy का न्यूनतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Completing squares: $(x-3)^2 + (y-3)^2 = 16$, centre $(3, 3)$, radius 4.

Parametric form: $x = 3 + 4\cos\theta$, $y = 3 + 4\sin\theta$, $\theta \in [0, 2\pi)$.

$xy = 9 + 12\cos\theta + 12\sin\theta + 16\sin\theta\cos\theta$.

Let $u = \cos\theta + \sin\theta = \sqrt{2}\sin\left(\theta + \dfrac{\pi}{4}\right)$, so u takes every value in $[-\sqrt{2}, \sqrt{2}]$, and $u^2 = 1 + 2\sin\theta\cos\theta$, so $\sin\theta\cos\theta = \dfrac{u^2 - 1}{2}$.

$xy = 9 + 12u + 8(u^2 - 1) = 8u^2 + 12u + 1$.

This quadratic opens upward with vertex at $u = -\dfrac{12}{16} = -\dfrac{3}{4}$, which lies in $[-\sqrt{2}, \sqrt{2}]$.

Minimum value $= 8 \cdot \dfrac{9}{16} + 12\left(-\dfrac{3}{4}\right) + 1 = \dfrac{9}{2} - 9 + 1 = -\dfrac{7}{2} = -3.5$.

Final Answer: -3.5

Question 11

English

Let P be the point of the curve $y = x^3$, $x > 0$, at which the tangent to the curve also touches the circle $x^2 + y^2 = \frac{1}{25}$. If this tangent touches the circle at the point (a, b), then the value of a is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि P वक्र $y = x^3$, $x > 0$ का वह बिन्दु है जिस पर वक्र की स्पर्श रेखा वृत्त $x^2 + y^2 = \frac{1}{25}$ को भी स्पर्श करती है। यदि यह स्पर्श रेखा वृत्त को बिन्दु (a, b) पर स्पर्श करती है, तो a का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

At (t, t^3) the slope of $y = x^3$ is $3t^2$, so the tangent is $y - t^3 = 3t^2(x - t)$, i.e. $3t^2x - y - 2t^3 = 0$.

It touches $x^2 + y^2 = \frac{1}{25}$ when its distance from the origin is $\frac{1}{5}$: $\frac{2t^3}{\sqrt{9t^4 + 1}} = \frac{1}{5}$ (with $t > 0$).

Squaring: $100t^6 = 9t^4 + 1$. With $u = t^2 > 0$: $100u^3 - 9u^2 - 1 = 0$.

$u = \frac{1}{4}$: $\frac{100}{64} - \frac{9}{16} - 1 = 0$, so $(4u - 1)$ is a factor: $100u^3 - 9u^2 - 1 = (4u - 1)(25u^2 + 4u + 1)$, and $25u^2 + 4u + 1$ has discriminant $16 - 100 < 0$.

So $u = \frac{1}{4}$, $t = \frac{1}{2}$ is the only point, $P = \left(\frac{1}{2}, \frac{1}{8}\right)$.

Tangent: $y = \frac{3}{4}x - \frac{1}{4}$, i.e. $3x - 4y - 1 = 0$; its distance from O is $\frac{1}{5}$, as required.

The radius to the point of contact is perpendicular to the tangent, so the point of contact is the foot of the perpendicular from O: $(a, b) = \lambda(3, -4)$ with $3(3\lambda) - 4(-4\lambda) - 1 = 0$, $\lambda = \frac{1}{25}$.

$(a, b) = \left(\frac{3}{25}, -\frac{4}{25}\right)$, so $a = 0.12$.

Final Answer: 0.12

Question 12

English

Consider the parabola $y^2 = 4x$ and the circle $C : x^2 + y^2 - 2x - 89 = 0$. The normals to the parabola that are also tangents to C form, together with the y-axis, a triangle. Then the area of this triangle is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

परवलय $y^2 = 4x$ तथा वृत्त $C : x^2 + y^2 - 2x - 89 = 0$ पर विचार कीजिए। परवलय के वे अभिलम्ब जो C की स्पर्श रेखाएँ भी हैं, y-अक्ष के साथ एक त्रिभुज बनाते हैं। तब इस त्रिभुज का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

C: $(x-1)^2 + y^2 = 90$, centre $(1, 0)$, which is the focus S of $y^2 = 4x$, radius $\sqrt{90}$.

The normal at $(t^2, 2t)$ is $y = -tx + 2t + t^3$, i.e. $tx + y - 2t - t^3 = 0$.

Distance from $S(1, 0)$: $\dfrac{|t - 2t - t^3|}{\sqrt{t^2+1}} = \dfrac{|t|(1+t^2)}{\sqrt{1+t^2}} = |t|\sqrt{1+t^2}$.

The normal touches C when this equals $\sqrt{90}$: $t^2(1+t^2) = 90$, so $t^4 + t^2 - 90 = 0$, $(t^2+10)(t^2-9) = 0$, $t^2 = 9$, $t = \pm 3$. (The normal $t = 0$ is the axis, which passes through the centre and is not a tangent.)

$t = 3$: $y = -3x + 33$; $t = -3$: $y = 3x - 33$.

Both meet the x-axis at $(11, 0)$. They meet the y-axis at $(0, 33)$ and $(0, -33)$.

The triangle has vertices $(11, 0)$, $(0, 33)$, $(0, -33)$: base 66 on the y-axis, height 11.

Area $= \dfrac{1}{2} \cdot 66 \cdot 11 = 363$.

Final Answer: 363

Question 13

English

Let E be the ellipse $\frac{x^2}{16}+\frac{y^2}{12}=1$, and let C be the circle with centre $K\left(\frac{1}{2},0\right)$ that touches E at two distinct points A and B. The tangents to E at A and B meet at T. Then the area of the quadrilateral $KATB$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि E दीर्घवृत्त $\frac{x^2}{16}+\frac{y^2}{12}=1$ है, तथा C केन्द्र $K\left(\frac{1}{2},0\right)$ वाला वह वृत्त है जो E को दो भिन्न बिन्दुओं A तथा B पर स्पर्श करता है। A तथा B पर E की स्पर्श रेखाएँ T पर मिलती हैं। तब चतुर्भुज $KATB$ का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

At a point where C touches E, the two curves have the same tangent, so they also have the same normal. The normal to C at A is the radius KA, so KA is the normal to E at A.

For E: $a^2=16$, $b^2=12$. Normal at $(4\cos\theta, 2\sqrt{3}\sin\theta)$: $\frac{ax}{\cos\theta}-\frac{by}{\sin\theta}=a^2-b^2$, i.e. $\frac{4x}{\cos\theta}-\frac{2\sqrt{3}y}{\sin\theta}=4$ (for $\sin\theta\neq 0$).

It passes through $K\left(\frac{1}{2},0\right)$: $\frac{2}{\cos\theta}=4$, so $\cos\theta=\frac{1}{2}$, $\sin\theta=\pm\frac{\sqrt{3}}{2}$, giving $A=(2,3)$, $B=(2,-3)$.

(The normals at the vertices $(\pm 4,0)$ are the x-axis, which also passes through K, but they give circles touching E at one point only.) Check: the squared distance from K to (x,y) on E is $\left(x-\frac{1}{2}\right)^2+12-\frac{3x^2}{4}=\frac{x^2}{4}-x+\frac{49}{4}$, least at $x=2$, so C (radius$^2=\frac{45}{4}$) lies inside E touching it at A and B.

Tangent to E at $(2,3)$: $\frac{2x}{16}+\frac{3y}{12}=1$, i.e. $x+2y=8$; at $(2,-3)$: $x-2y=8$. They meet at $T=(8,0)$.

$KA\perp AT$ and $KB\perp BT$ (radius perpendicular to tangent), and $KATB$ is symmetric about the x-axis: a kite with diagonals $KT=8-\frac{1}{2}=\frac{15}{2}$ and $AB=6$, which are perpendicular.

Area $=\frac{1}{2}\cdot\frac{15}{2}\cdot 6=\frac{45}{2}=22.5$.

Final Answer: 22.5

Question 14

English

Let C be the circle $x^2 + y^2 - 2x - 4y - 4 = 0$ and $P = (4, 6)$. The tangents drawn from P to C touch C at A and B. The tangent to C at the point of the minor arc AB nearest to P meets PA and PB at D and E, respectively. Then the area of the quadrilateral $ABED$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि C वृत्त $x^2 + y^2 - 2x - 4y - 4 = 0$ है तथा $P = (4, 6)$ है। P से C पर खींची गई स्पर्श रेखाएँ C को A तथा B पर स्पर्श करती हैं। लघु चाप AB के P के निकटतम बिन्दु पर C की स्पर्श रेखा PA तथा PB को क्रमशः D तथा E पर मिलती है। तब चतुर्भुज $ABED$ का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

C: $(x-1)^2 + (y-2)^2 = 9$, centre $O = (1, 2)$, radius 3. $OP = \sqrt{9+16} = 5 > 3$, so P is outside C.
Length of tangent: $PA = PB = \sqrt{S(P)} = \sqrt{16 + 36 - 8 - 24 - 4} = 4$ (also $\sqrt{OP^2 - r^2}$).
Let $\alpha = \angle APO$. In right triangle OAP: $\sin\alpha = \dfrac{3}{5}$, $\cos\alpha = \dfrac{4}{5}$.
AB is the chord of contact of P; it is perpendicular to OP. Its distance from P is $PA\cos\alpha = \dfrac{16}{5}$ and its half-length is $PA\sin\alpha = \dfrac{12}{5}$, so area of triangle $PAB = \dfrac{16}{5} \cdot \dfrac{12}{5} = \dfrac{192}{25} = 7.68$.
The point of the minor arc nearest to P is Q, where OP meets C: $PQ = OP - r = 2$.
The tangent at Q is perpendicular to OQ, i.e. to OP, so it is parallel to AB. It cuts PA, PB at D, E with $QD = QE = PQ\tan\alpha = 2 \cdot \dfrac{3}{4} = \dfrac{3}{2}$.
Area of triangle $PDE = \dfrac{1}{2} \cdot DE \cdot PQ = \dfrac{1}{2} \cdot 3 \cdot 2 = 3$.
$ABED$ is the trapezium between the parallel lines AB and DE: area $= 7.68 - 3 = 4.68$.

Final Answer: 4.68

Question 15

English

Let $C_1 : x^2 + y^2 = 4$ and $C_2 : (x-2)^2 + (y-1)^2 = 4$. Let S be the set of all points P on the line $2x + y + 1 = 0$ such that P lies outside C_1 and the chord of contact of the tangents drawn from P to C_1 touches C_2. Then the sum of the abscissae of all the points of S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $C_1 : x^2 + y^2 = 4$ तथा $C_2 : (x-2)^2 + (y-1)^2 = 4$ हैं। माना S रेखा $2x + y + 1 = 0$ पर स्थित उन सभी बिन्दुओं P का समुच्चय है जिनके लिए P, C_1 के बाहर स्थित है तथा P से C_1 पर खींची गई स्पर्श रेखाओं की स्पर्श जीवा C_2 को स्पर्श करती है। तब S के सभी बिन्दुओं के भुजों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A point of the line is $P = (h, k)$ with $k = -2h - 1$.

For P outside C_1 the chord of contact is $T = 0$: $hx + ky = 4$.

C_2 has centre $(2, 1)$ and radius 2. The chord of contact touches C_2 when its distance from $(2, 1)$ is 2: $\dfrac{|2h + k - 4|}{\sqrt{h^2 + k^2}} = 2$.

With $k = -2h - 1$: $2h + k - 4 = -5$, and $h^2 + k^2 = 5h^2 + 4h + 1$. So $25 = 4(5h^2 + 4h + 1)$, i.e. $20h^2 + 16h - 21 = 0$.

Discriminant $256 + 1680 = 1936 = 44^2$: $h = \dfrac{-16 \pm 44}{40}$, so $h = \dfrac{7}{10}$ or $h = -\dfrac{3}{2}$.

Points: $\left(\dfrac{7}{10}, -\dfrac{12}{5}\right)$ and $\left(-\dfrac{3}{2}, 2\right)$. For both, $h^2 + k^2 = 5h^2 + 4h + 1 = \dfrac{25}{4} > 4$, so both lie outside C_1 and the tangents exist.

Sum of abscissae $= \dfrac{7}{10} - \dfrac{3}{2} = -\dfrac{4}{5} = -0.8$.

Final Answer: -0.8

Question 16

English

Let C be the circle $x^2 + y^2 - 2x + 4y - 95 = 0$. A chord AB of C has its midpoint at $M(1, 6)$. The tangents to C at A and B meet at P. Then the area of the triangle PAB is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि C वृत्त $x^2 + y^2 - 2x + 4y - 95 = 0$ है। C की एक जीवा AB का मध्यबिन्दु $M(1, 6)$ है। A तथा B पर C की स्पर्श रेखाएँ P पर मिलती हैं। तब त्रिभुज PAB का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

C: $(x-1)^2 + (y+2)^2 = 100$, centre $O(1, -2)$, radius 10. $M(1, 6)$ is inside, $OM = 8$.

The chord with midpoint M is perpendicular to OM (vertical), so it is the line $y = 6$, with half-length $\sqrt{100 - 64} = 6$: $A = (-5, 6)$, $B = (7, 6)$, $AB = 12$.

The tangents at A and B meet at P, and AB is the chord of contact of P. By symmetry P is on line OM ($x = 1$).

In right triangle OAP (right angle at A), AM is the altitude, so $OA^2 = OM \cdot OP$: $OP = \dfrac{100}{8} = \dfrac{25}{2}$.

So $P = \left(1, -2 + \dfrac{25}{2}\right) = \left(1, \dfrac{21}{2}\right)$. Check: the chord of contact of P is $x \cdot 1 + y \cdot \dfrac{21}{2} - (x + 1) + 2\left(y + \dfrac{21}{2}\right) - 95 = 0$, i.e. $\dfrac{25}{2}y = 75$, $y = 6$.

$PM = \dfrac{21}{2} - 6 = \dfrac{9}{2}$.

Area of triangle $PAB = \dfrac{1}{2} \cdot 12 \cdot \dfrac{9}{2} = 27$.

Final Answer: 27

Question 17

English

Let C be the circle $x^2 + y^2 = 100$ and $P = (1, 7)$. Exactly two chords of C that pass through P have their midpoints on the line $y = x + 3$. If ℓ_1 and ℓ_2 are the lengths of these two chords, then the value of $\ell_1^2 + \ell_2^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि C वृत्त $x^2 + y^2 = 100$ है तथा $P = (1, 7)$ है। C की ठीक दो जीवाएँ, जो P से होकर जाती हैं, के मध्यबिन्दु रेखा $y = x + 3$ पर स्थित हैं। यदि ℓ_1 तथा ℓ_2 इन दो जीवाओं की लम्बाइयाँ हैं, तो $\ell_1^2 + \ell_2^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The chord of $x^2 + y^2 = 100$ with midpoint $M(h, k)$ is $T = S_1$: $hx + ky = h^2 + k^2$.

It passes through $P(1, 7)$ iff $h + 7k = h^2 + k^2$. Hence the midpoints of chords through P lie on $x^2 + y^2 - x - 7y = 0$, the circle with OP as diameter ($OM \perp MP$). P is inside C ($1 + 49 < 100$), so the whole circle is traced.

Midpoints on $y = x + 3$: $x^2 + (x + 3)^2 - x - 7(x + 3) = 0$, i.e. $2x^2 - 2x - 12 = 0$, $x^2 - x - 6 = 0$, $x = 3$ or $x = -2$.

Midpoints $M_1 = (3, 6)$ and $M_2 = (-2, 1)$, neither equal to O or P, so the two chords are well defined (chords $3x + 6y = 45$ and $-2x + y = 5$, both through P).

Half-length of a chord with midpoint M is $\sqrt{100 - OM^2}$, so $\ell^2 = 4(100 - OM^2)$.

$OM_1^2 = 45$: $\ell_1^2 = 220$. $OM_2^2 = 5$: $\ell_2^2 = 380$.

$\ell_1^2 + \ell_2^2 = 600$.

Final Answer: 600

Question 18

English

Consider the parabola $y^2 = -8x$ with focus F and the circle $C : x^2 + y^2 + 14x + 25 = 0$. Let $\mathcal{L}$ be the set of all lines through F, each of which meets the parabola at two distinct points and meets C at two distinct points, such that the chord of the parabola and the chord of C cut off by the line have the same midpoint. Then the product of the y-intercepts of the lines in $\mathcal{L}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

नाभि F वाले परवलय $y^2 = -8x$ तथा वृत्त $C : x^2 + y^2 + 14x + 25 = 0$ पर विचार कीजिए। माना $\mathcal{L}$, F से होकर जाने वाली उन सभी रेखाओं का समुच्चय है जिनमें से प्रत्येक परवलय को दो भिन्न बिन्दुओं पर तथा C को दो भिन्न बिन्दुओं पर मिलती है, तथा रेखा द्वारा काटी गई परवलय की जीवा तथा C की जीवा का मध्यबिन्दु एक ही है। तब $\mathcal{L}$ की रेखाओं के y-अन्तःखण्डों का गुणनफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$y^2 = -8x$ has $4a = 8$, focus $F = (-2, 0)$. C: $(x + 7)^2 + y^2 = 24$, centre $O' = (-7, 0)$, radius $\sqrt{24}$.

Let the common midpoint be $M(h, k)$.

Parabola: the chord with midpoint M is $T = S_1$: $ky + 4(x + h) = k^2 + 8h$, i.e. $ky + 4x = k^2 + 4h$. For $k \neq 0$ its slope is $-\dfrac{4}{k}$.

Circle: the chord with midpoint M is perpendicular to $O'M$, whose slope is $\dfrac{k}{h+7}$. So $\dfrac{k}{h+7} \cdot \left(-\dfrac{4}{k}\right) = -1$, giving $h + 7 = 4$, $h = -3$.

The line passes through $F(-2, 0)$: slope of $FM = \dfrac{k}{h+2} = \dfrac{k}{-1}$ must equal $-\dfrac{4}{k}$, so $k^2 = 4$, $k = \pm 2$.

$k = 2$: slope -2, line $y = -2(x + 2)$; $k = -2$: slope 2, line $y = 2(x + 2)$.

Case $k = 0$: the chord is vertical, $x = -2$ through F; its distance from O' is $5 > \sqrt{24}$, so it does not meet C and is excluded.

Existence: $M = (-3, \pm 2)$ satisfies $k^2 = 4 < -8h = 24$, so M is inside the parabola, and $(h + 7)^2 + k^2 = 20 < 24$, so M is inside C. Each line therefore cuts both curves in chords bisected at M.

y-intercepts: -4 and 4. Product $= -16$.

Final Answer: -16

Question 19

English

Let $P = \left(-1, \frac{3}{2}\right)$. The chord of contact of the tangents drawn from P to the parabola $y^2 = 4x$ is L_1, and the chord of contact of the tangents drawn from P to the circle $(x-1)^2 + y^2 = 1$ is L_2. Then the distance between L_1 and L_2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $P = \left(-1, \frac{3}{2}\right)$ है। P से परवलय $y^2 = 4x$ पर खींची गई स्पर्श रेखाओं की स्पर्श जीवा L_1 है, तथा P से वृत्त $(x-1)^2 + y^2 = 1$ पर खींची गई स्पर्श रेखाओं की स्पर्श जीवा L_2 है। तब L_1 तथा L_2 के बीच की दूरी _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$P(-1, \frac{3}{2})$ lies on the directrix $x = -1$ of $y^2 = 4x$, and outside both curves, so both chords of contact exist.

Chord of contact with respect to $y^2 = 4x$ ($T = 0$): $y \cdot \frac{3}{2} = 2(x-1)$, i.e. $4x - 3y - 4 = 0$. It passes through the focus $S(1, 0)$.

The circle $(x-1)^2 + y^2 = 1$ is centred at the focus. P is outside it: $4 + \frac{9}{4} > 1$.

Chord of contact with respect to the circle ($T = 0$): $(x-1)(-1-1) + y \cdot \frac{3}{2} = 1$, i.e. $-2x + 2 + \frac{3}{2}y = 1$, i.e. $4x - 3y - 2 = 0$.

The two lines have the same normal $(4, -3)$, parallel to $\overrightarrow{SP} = (-2, \frac{3}{2})$, so they are parallel.

Distance $= \frac{|-4-(-2)|}{\sqrt{16+9}} = \frac{2}{5} = 0.4$. (In general it equals $\frac{r^2}{SP} = \frac{1}{5/2}$.)

Final Answer: 0.4

Question 20

English

For a non-zero real number k, let C_k be the circle $x^2 + y^2 + 2kx = 0$, and let C be the circle $(x-4)^2 + y^2 = 4$. Then the sum of all values of k for which C_k and C touch each other is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

किसी शून्येतर वास्तविक संख्या k के लिए माना कि C_k वृत्त $x^2 + y^2 + 2kx = 0$ है, तथा C वृत्त $(x-4)^2 + y^2 = 4$ है। तब k के उन सभी मानों का योग, जिनके लिए C_k तथा C एक-दूसरे को स्पर्श करते हैं, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

C_k: $(x+k)^2 + y^2 = k^2$, centre $(-k, 0)$, radius $|k|$ (it passes through the origin). C: centre $(4, 0)$, radius 2. Distance between centres $d = |4 + k|$.

Two circles touch externally (then they have 3 common tangents) when $d = r_1 + r_2$, and internally (then 1 common tangent) when $d = |r_1 - r_2|$.

External: $|4 + k| = |k| + 2$.

$k > 0$: $4 + k = k + 2$, impossible. $-4 \le k < 0$: $4 + k = -k + 2$, $k = -1$ (valid). $k < -4$: $-4 - k = -k + 2$, impossible.

Internal: $|4 + k| = \big||k| - 2\big|$.

$k > 0$: $4 + k = |k - 2|$; for $k \ge 2$ this gives $4 = -2$, for $0 < k < 2$ it gives $k = -1$, not in range. $-4 \le k < 0$: $4 + k = |-k - 2| = |k + 2|$; for $-2 \le k < 0$: $4 = 2$, impossible; for $-4 \le k < -2$: $4 + k = -k - 2$, $k = -3$ (valid). $k < -4$: $-4 - k = -k - 2$, impossible.

Check: $k = -1$: centres $(1, 0)$, $(4, 0)$, radii 1, 2, $d = 3 = 1 + 2$. $k = -3$: centres $(3, 0)$, $(4, 0)$, radii 3, 2, $d = 1 = 3 - 2$.

Sum $= -1 + (-3) = -4$.

Final Answer: -4

Question 21

English

Let $C_1 : x^2 + y^2 - 2x - 4y - 4 = 0$ and $C_2 : x^2 + y^2 - 6x + 2y + 9 = 0$. All the common tangents of C_1 and C_2 pass through a point (α, β). Then the value of β is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $C_1 : x^2 + y^2 - 2x - 4y - 4 = 0$ तथा $C_2 : x^2 + y^2 - 6x + 2y + 9 = 0$ हैं। C_1 तथा C_2 की सभी उभयनिष्ठ स्पर्श रेखाएँ एक बिन्दु (α, β) से होकर जाती हैं। तब β का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

C_1: $(x-1)^2 + (y-2)^2 = 9$, centre $(1, 2)$, radius 3. C_2: $(x-3)^2 + (y+1)^2 = 1$, centre $(3, -1)$, radius 1.

Distance between centres $d = \sqrt{4+9} = \sqrt{13} \approx 3.61$.

Since $|r_1 - r_2| = 2 < d < 4 = r_1 + r_2$, the circles cut at two points. Then there are exactly two common tangents and both are direct (transverse tangents would have to pass between the circles, which is impossible when they overlap).

Two direct common tangents of circles with unequal radii meet on the line of centres at the external centre of similitude, which divides C_1C_2 externally in the ratio $r_1 : r_2 = 3 : 1$.

$$(\alpha, \beta) = \left(\frac{3 \cdot 3 - 1 \cdot 1}{3 - 1}, \frac{3(-1) - 1 \cdot 2}{3 - 1}\right) = \left(4, -\frac{5}{2}\right).$$

Check: the distances of $(4, -\frac{5}{2})$ from the centres are $\sqrt{29.25}$ and $\sqrt{3.25}$, in ratio $3 : 1$.

$\beta = -2.5$.

Final Answer: -2.5

Question 22

English

Two circles C_1 and C_2 lie outside each other. The radius of C_1 is 4. The length of a direct common tangent of C_1 and C_2 is 15 and the length of a transverse common tangent is 9. If d is the distance between the centres of C_1 and C_2, then the value of d^2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

दो वृत्त C_1 तथा C_2 एक-दूसरे के बाहर स्थित हैं। C_1 की त्रिज्या 4 है। C_1 तथा C_2 की एक अनुस्पर्श (direct) उभयनिष्ठ स्पर्श रेखा की लम्बाई 15 है तथा एक तिर्यक (transverse) उभयनिष्ठ स्पर्श रेखा की लम्बाई 9 है। यदि d दोनों वृत्तों के केन्द्रों के बीच की दूरी है, तो d^2 का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For circles with radii r_1, r_2 and centres d apart (outside each other), the lengths of the direct and transverse common tangents are $L_d = \sqrt{d^2 - (r_1 - r_2)^2}$ and $L_t = \sqrt{d^2 - (r_1 + r_2)^2}$.

Then $L_d^2 - L_t^2 = (r_1 + r_2)^2 - (r_1 - r_2)^2 = 4r_1r_2$, independent of d.

$225 - 81 = 144 = 4 \cdot 4 \cdot r_2$, so $r_2 = 9$.

From the transverse tangent: $d^2 = 81 + (4 + 9)^2 = 81 + 169 = 250$.

Check with the direct tangent: $d^2 = 225 + (9 - 4)^2 = 250$. Also $d = \sqrt{250} > 13 = r_1 + r_2$, so the circles are indeed outside each other.

Final Answer: 250

Question 23

English

Let S be the set of all points P on the line $y = x$ for which the polar of P with respect to the circle $x^2 + y^2 = 1$ touches the circle $(x - 3)^2 + y^2 = \frac{1}{2}$. Then the sum of the abscissae of all the points of S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S रेखा $y = x$ पर स्थित उन सभी बिन्दुओं P का समुच्चय है जिनके लिए वृत्त $x^2 + y^2 = 1$ के सापेक्ष P की ध्रुवी, वृत्त $(x - 3)^2 + y^2 = \frac{1}{2}$ को स्पर्श करती है। तब S के सभी बिन्दुओं के भुजों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A point of the line is $P = (p, p)$. For $p = 0$ the polar does not exist (the centre has no polar), so $p \neq 0$.

The polar of (p, p) with respect to $x^2 + y^2 = 1$ is $T = 0$: $px + py = 1$, i.e. $x + y = \frac{1}{p}$.

It touches $(x-3)^2+y^2 = \frac{1}{2}$ when its distance from $(3, 0)$ equals $\frac{1}{\sqrt{2}}$: $\frac{\left|3 - \frac{1}{p}\right|}{\sqrt{2}} = \frac{1}{\sqrt{2}}$, so $\left|3 - \frac{1}{p}\right| = 1$.

$\frac{1}{p} = 2$ or $\frac{1}{p} = 4$, i.e. $p = \frac{1}{2}$ or $p = \frac{1}{4}$.

Both points $\left(\frac{1}{2}, \frac{1}{2}\right)$ and $\left(\frac{1}{4}, \frac{1}{4}\right)$ lie inside $x^2 + y^2 = 1$; the polar is defined for them (it is not a chord of contact here, which the question does not need).

Sum of abscissae $= \dfrac{3}{4} = 0.75$.

Final Answer: 0.75

Question 24

English

Let $S_1 : x^2 + y^2 - 4 = 0$, $S_2 : x^2 + y^2 - 2x + 4y - 6 = 0$ and $S_3 : x^2 + y^2 - 6x - 2y + 6 = 0$. Among the circles passing through the points of intersection of $S_1 = 0$ and $S_2 = 0$, let Γ be the one that bisects the circumference of $S_3 = 0$. If ρ is the radius of Γ, then the value of ρ^2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $S_1 : x^2 + y^2 - 4 = 0$, $S_2 : x^2 + y^2 - 2x + 4y - 6 = 0$ तथा $S_3 : x^2 + y^2 - 6x - 2y + 6 = 0$ हैं। $S_1 = 0$ तथा $S_2 = 0$ के प्रतिच्छेद बिन्दुओं से होकर जाने वाले वृत्तों में से माना Γ वह वृत्त है जो $S_3 = 0$ की परिधि को समद्विभाजित करता है। यदि ρ, Γ की त्रिज्या है, तो ρ^2 का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Every circle through the common points of $S_1 = 0$ and $S_2 = 0$ (other than $S_2 = 0$) is $S_1 + \lambda S_2 = 0$, $\lambda \neq -1$. Dividing by $1 + \lambda$ makes the coefficients of x^2, y^2 equal to 1: $x^2 + y^2 + \dfrac{-2\lambda x + 4\lambda y - (4 + 6\lambda)}{1 + \lambda} = 0$.

S_3: $(x - 3)^2 + (y - 1)^2 = 4$, centre $(3, 1)$, radius 2.

A circle bisects the circumference of S_3 when their common chord is a diameter of S_3, i.e. the radical axis $\Gamma - S_3 = 0$ (both in normalised form) passes through $(3, 1)$.

At $(3, 1)$: $\Gamma = 9+1+\dfrac{-6\lambda + 4\lambda - 4 - 6\lambda}{1 + \lambda} = 10+\dfrac{-8\lambda - 4}{1 + \lambda} = \dfrac{2\lambda + 6}{1 + \lambda}$, and $S_3(3, 1) = 9+1-18-2+6 = -4$.

Condition $\dfrac{2\lambda + 6}{1 + \lambda} = -4$: $2\lambda + 6 = -4 - 4\lambda$, $\lambda = -\dfrac{5}{3}$.

Then $1 + \lambda = -\dfrac{2}{3}$ and Γ: $x^2 + y^2 + \dfrac{\frac{10}{3}x - \frac{20}{3}y + 6}{-\frac{2}{3}} = 0$, i.e. $x^2 + y^2 - 5x + 10y - 9 = 0$.

Check: $\Gamma - S_3 = x + 12y - 15$, which passes through $(3, 1)$. Since it passes through the centre, it meets S_3 in two points, which lie on Γ too.

$\rho^2 = \left(\dfrac{5}{2}\right)^2 + 5^2 + 9 = \dfrac{25}{4} + 34 = 40.25$.

Final Answer: 40.25

Question 25

English

Let $C_1 : x^2 + y^2 = 25$, $C_2 : x^2 + y^2 - 6x - 19 = 0$ and $C_3 : x^2 + y^2 - 4y - 17 = 0$. The common chord of C_1 and C_2 meets C_1 at A and B, and it intersects the common chord of C_2 and C_3 at the point K. Then the value of $KA \cdot KB$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $C_1 : x^2 + y^2 = 25$, $C_2 : x^2 + y^2 - 6x - 19 = 0$ तथा $C_3 : x^2 + y^2 - 4y - 17 = 0$ हैं। C_1 तथा C_2 की उभयनिष्ठ जीवा C_1 को A तथा B पर मिलती है, तथा यह C_2 तथा C_3 की उभयनिष्ठ जीवा को बिन्दु K पर प्रतिच्छेद करती है। तब $KA \cdot KB$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Common chord of C_1 and C_2 (their radical axis): $S_1 - S_2 = 6x - 6 = 0$, i.e. $x = 1$. It cuts C_1 where $y^2 = 24$, so the circles do intersect, at $A = (1, 2\sqrt{6})$ and $B = (1, -2\sqrt{6})$; $AB = 4\sqrt{6}$.

Common chord of C_2 and C_3: $S_2 - S_3 = -6x + 4y - 2 = 0$, i.e. $3x - 2y + 1 = 0$. (C_2: centre $(3, 0)$, $r^2 = 28$; C_3: centre $(0, 2)$, $r^2 = 21$; $d^2 = 13$, so they intersect.)

K: $x = 1$ and $3 - 2y + 1 = 0$, so $K = (1, 2)$. K lies on all three radical axes, so it is the radical centre: $S_1(K) = S_2(K) = S_3(K)$.

$S_1(K) = 1 + 4 - 25 = -20$ (also $S_2(K) = 1 + 4 - 6 - 19 = -20$), so K is inside C_1 and lies between A and B.

For a chord AB of C_1 through the interior point K, $KA \cdot KB = r^2 - OK^2 = -S_1(K) = 20$.

Directly: $KA = 2\sqrt{6} - 2$, $KB = 2\sqrt{6} + 2$, so $KA \cdot KB = 24 - 4 = 20$.

Final Answer: 20

Question 26

English

The line $L : x = 12$ meets the circle $C_1 : x^2 + y^2 = 169$ at the point P in the first quadrant, and θ is the acute angle at which L cuts C_1 at P. Let C_2 be a circle of radius 7 whose centre lies on L and which passes through P, and let ϕ be the acute angle between C_1 and C_2 at P. Then the value of $\dfrac{\tan\phi}{\tan\theta}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

रेखा $L : x = 12$ वृत्त $C_1 : x^2 + y^2 = 169$ को प्रथम चतुर्थांश के बिन्दु P पर मिलती है, तथा θ वह न्यून कोण है जिस पर L, C_1 को P पर काटती है। माना C_2 त्रिज्या 7 का एक वृत्त है जिसका केन्द्र L पर स्थित है तथा जो P से होकर जाता है, तथा ϕ बिन्दु P पर C_1 तथा C_2 के बीच का न्यून कोण है। तब $\dfrac{\tan\phi}{\tan\theta}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$x = 12$ on $x^2 + y^2 = 169$ gives $y = \pm 5$; in the first quadrant $P = (12, 5)$.

The angle at which a line cuts a circle is the angle between the line and the tangent at the point of intersection. The tangent to C_1 at P is $12x + 5y = 169$, with direction $(5, -12)$; L has direction $(0, 1)$.

$\cos\theta = \dfrac{|0 \cdot 5 + 1 \cdot (-12)|}{13} = \dfrac{12}{13}$, so $\sin\theta = \dfrac{5}{13}$ and $\tan\theta = \dfrac{5}{12}$.

C_2 has its centre on L and passes through $P \in L$, so its radius at P lies along L. Its tangent at P is perpendicular to L.

The angle between C_1 and C_2 is the angle between their tangents at P. Since the tangent of C_2 is perpendicular to L, this angle is $\phi = \dfrac{\pi}{2} - \theta$, so $\tan\phi = \cot\theta = \dfrac{12}{5}$ (the same for both possible centres $(12, 12)$ and $(12, -2)$).

Check with the formula $\cos\phi = \dfrac{|r_1^2 + r_2^2 - d^2|}{2r_1 r_2}$: centre $(12, 12)$, $d^2 = 288$: $\dfrac{|169 + 49 - 288|}{182} = \dfrac{70}{182} = \dfrac{5}{13}$; centre $(12, -2)$, $d^2 = 148$: $\dfrac{70}{182} = \dfrac{5}{13}$. So $\tan\phi = \dfrac{12}{5}$.

$\dfrac{\tan\phi}{\tan\theta} = \dfrac{12}{5} \cdot \dfrac{12}{5} = \dfrac{144}{25} = 5.76$.

Final Answer: 5.76

Question 27

English

A circle C with its centre on the negative x-axis passes through the point $(0, 5)$ and cuts the circle $x^2 + y^2 = 25$ at an angle θ such that $\tan\theta = \frac{3}{4}$. If the centre of C is $(\alpha, 0)$, then the value of α is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक वृत्त C, जिसका केन्द्र ऋणात्मक x-अक्ष पर है, बिन्दु $(0, 5)$ से होकर जाता है तथा वृत्त $x^2 + y^2 = 25$ को कोण θ पर काटता है, जहाँ $\tan\theta = \frac{3}{4}$ है। यदि C का केन्द्र $(\alpha, 0)$ है, तो α का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

C passes through $(0, 5)$, which also lies on $x^2 + y^2 = 25$, so $(0, 5)$ is a common point. Radius of C: $r = \sqrt{\alpha^2 + 25}$.

The angle between two circles at a common point is the angle between their tangents there, equivalently between the radii: $\cos\theta = \dfrac{|r_1^2 + r_2^2 - d^2|}{2r_1 r_2}$.

Here $r_1 = 5$, $r_2 = r$, $d = |\alpha|$: $\cos\theta = \dfrac{25 + \alpha^2 + 25 - \alpha^2}{10r} = \dfrac{5}{r}$.

$\tan\theta = \frac{3}{4}$ with θ acute gives $\cos\theta = \frac{4}{5}$, so $r = \frac{25}{4}$.

$\alpha^2 = r^2 - 25 = \frac{625}{16} - \frac{400}{16} = \frac{225}{16}$, $\alpha = \pm\frac{15}{4}$.

The centre is on the negative x-axis, so $\alpha = -\frac{15}{4} = -3.75$. (Check: $d = 3.75$, and $|5 - 6.25| < 3.75 < 11.25$, so the circles do intersect.)

Final Answer: -3.75

Question 28

English

Let Γ be the circle that cuts the circle $x^2 + y^2 = 9$ orthogonally, passes through the point $(1, 0)$ and has its centre on the line $x + y = 2$. Then the y-coordinate of the lowest point of Γ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि Γ वह वृत्त है जो वृत्त $x^2 + y^2 = 9$ को लम्बकोणीय रूप से काटता है, बिन्दु $(1, 0)$ से होकर जाता है तथा जिसका केन्द्र रेखा $x + y = 2$ पर है। तब Γ के सबसे निचले बिन्दु का y-निर्देशांक _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let Γ: $x^2 + y^2 + 2gx + 2fy + c = 0$.

Orthogonality with $x^2 + y^2 - 9 = 0$ ($g_1 = f_1 = 0$, $c_1 = -9$): $2(gg_1 + ff_1) = c + c_1$, i.e. $0 = c - 9$, so $c = 9$.

Through $(1, 0)$: $1 + 2g + 9 = 0$, so $g = -5$. (Every circle orthogonal to $x^2 + y^2 = 9$ through $(1, 0)$ has centre on $x = 5$ and also passes through $(9, 0)$, since $1 \cdot 9 = 3^2$.)

Centre $(-g, -f) = (5, -f)$ lies on $x + y = 2$: $5 - f = 2$, $f = 3$.

Γ: $x^2 + y^2 - 10x + 6y + 9 = 0$, centre $(5, -3)$, radius $\sqrt{25 + 9 - 9} = 5$. Check: distance of centres $\sqrt{34}$, and $34 = 9 + 25$.

The lowest point is $(5, -3 - 5) = (5, -8)$, so the required y-coordinate is -8.

Final Answer: -8

Question 29

English

Among all circles passing through the points $(1, 0)$ and $(5, 0)$, let Γ be the one that cuts the circle $C : x^2 + y^2 - 6x - 4y - 11 = 0$ orthogonally. If Γ and C intersect at P and Q, then the value of PQ^2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

बिन्दुओं $(1, 0)$ तथा $(5, 0)$ से होकर जाने वाले सभी वृत्तों में से माना Γ वह वृत्त है जो वृत्त $C : x^2 + y^2 - 6x - 4y - 11 = 0$ को लम्बकोणीय रूप से काटता है। यदि Γ तथा C बिन्दुओं P तथा Q पर प्रतिच्छेद करते हैं, तो PQ^2 का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Every circle through $(1, 0)$ and $(5, 0)$ belongs to the family $(x - 1)(x - 5) + y^2 + \lambda y = 0$ (the point circle on the chord plus λ times the line $y = 0$): $x^2 + y^2 - 6x + \lambda y + 5 = 0$, centre $\left(3, -\dfrac{\lambda}{2}\right)$.

C: $g = -3$, $f = -2$, $c = -11$; centre $(3, 2)$, radius$^2 = 9 + 4 + 11 = 24$.

Orthogonality: $2(g_1g_2 + f_1f_2) = c_1 + c_2$: $2(9 - \lambda) = 5 - 11 = -6$, so $\lambda = 12$.

Γ: $x^2 + y^2 - 6x + 12y + 5 = 0$, centre $(3, -6)$, radius$^2 = 9 + 36 - 5 = 40$. Distance of centres $d = 8$; check $64 = 40 + 24$.

The common chord PQ is perpendicular to the line of centres. In the right triangle formed by the two centres and P (right angle at P, by orthogonality), the half-chord is the altitude to the

hypotenuse: $\dfrac{PQ}{2} = \dfrac{r_1 r_2}{d}$.

$PQ^2 = \dfrac{4r_1^2 r_2^2}{d^2} = \dfrac{4 \cdot 40 \cdot 24}{64} = 60.$

Check: $\Gamma - C = 16y + 16 = 0$, $y = -1$; on C: $x^2 - 6x - 6 = 0$, $x = 3 \pm \sqrt{15}$, so $PQ = 2\sqrt{15}$, $PQ^2 = 60$.

Final Answer: 60

Question 30

English

Let C be the circle $(x-5)^2 + y^2 = 9$, and let $m \in (0, 1)$. The image of C in the line $y = mx$ touches the y-axis. Then the value of m is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि C वृत्त $(x-5)^2 + y^2 = 9$ है तथा $m \in (0, 1)$ है। रेखा $y = mx$ में C का प्रतिबिम्ब y-अक्ष को स्पर्श करता है। तब m का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Reflection in a line maps a circle to a circle of the same radius whose centre is the image of the original centre.

Write $m = \tan\alpha$. For $m \in (0, 1)$, $0 < \alpha < \dfrac{\pi}{4}$. Reflecting $(5, 0)$ in the line through the origin at angle α gives the point at angle 2α: $(5\cos 2\alpha, 5\sin 2\alpha)$.

The image circle has radius 3. It touches the y-axis (tangent condition) when the distance of its centre from the y-axis equals 3: $|5\cos 2\alpha| = 3$.

Since $0 < 2\alpha < \dfrac{\pi}{2}$, $\cos 2\alpha > 0$, so $\cos 2\alpha = \dfrac{3}{5}$.

$\cos 2\alpha = \dfrac{1 - m^2}{1 + m^2} = \dfrac{3}{5}$ gives $5 - 5m^2 = 3 + 3m^2$, $m^2 = \dfrac{1}{4}$, $m = \dfrac{1}{2}$ (as $m > 0$).

Check: the image of $(5, 0)$ in $y = \dfrac{x}{2}$ is $(3, 4)$, at distance 3 from the y-axis.

Final Answer: 0.5

Question 31

English

Let C be the circle $x^2 + y^2 - 6x - 8y - 75 = 0$. For $k = 1, 2, \ldots, 20$, let M_k and m_k denote the greatest and the least distances, respectively, of the point $P_k(k, 4)$ from the points of C. Then the value of $\sum_{k=1}^{20} (M_k - m_k)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि C वृत्त $x^2 + y^2 - 6x - 8y - 75 = 0$ है। $k = 1, 2, \ldots, 20$ के लिए माना M_k तथा m_k क्रमशः बिन्दु $P_k(k, 4)$ की C के बिन्दुओं से अधिकतम तथा न्यूनतम दूरियाँ हैं। तब $\sum_{k=1}^{20} (M_k - m_k)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

C: $(x-3)^2 + (y-4)^2 = 100$, centre $O(3, 4)$, radius 10. All P_k lie on the line $y = 4$ through O, so $d_k = OP_k = |k - 3|$.

Position: P_k is inside or on C when $d_k \le 10$ ($k \le 13$) and outside when $k \ge 14$.

For any point at distance d from O, the greatest distance to the circle is $d + r$ (along PO produced). The least distance is $r - d$ if the point is inside or on C and $d - r$ if it is outside.

Hence $M_k - m_k = 2d_k$ for $k \le 13$ and $M_k - m_k = 2r = 20$ for $k \ge 14$. (At $k = 3$, $P_3 = O$ and $M_3 = m_3 = 10$.)

$k = 1$ to 13: $d_k = 2, 1, 0, 1, 2, \ldots, 10$, whose sum is $2 + 1 + 0 + 55 = 58$; contribution 116.

$k = 14$ to 20: 7 terms, each 20; contribution 140.

$$\sum_{k=1}^{20} (M_k - m_k) = 116 + 140 = 256.$$

Final Answer: 256

Question 32

English

Let C be the circle $x^2 + y^2 - 20x - 20y + 191 = 0$ and let H be the curve $y = \dfrac{18}{x}$, $x > 0$. Then the minimum value of PQ^2, where P is a point of C and Q is a point of H, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि C वृत्त $x^2 + y^2 - 20x - 20y + 191 = 0$ है तथा H वक्र $y = \dfrac{18}{x}$, $x > 0$ है। तब PQ^2 का न्यूनतम मान, जहाँ P, C का एक बिन्दु है तथा Q, H का एक बिन्दु है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

C: $(x-10)^2 + (y-10)^2 = 9$, centre $O(10, 10)$, radius 3.

For a fixed Q outside C, the least distance from Q to C is $OQ - 3$. So the minimum of PQ is $(\min_{Q \in H} OQ) - 3$, provided this is positive.

$Q = \left(t, \dfrac{18}{t}\right)$, $t > 0$: $OQ^2 = (t-10)^2 + \left(\dfrac{18}{t} - 10\right)^2 = t^2 + \dfrac{324}{t^2} - 20\left(t + \dfrac{18}{t}\right) + 200$.

Let $u = t + \dfrac{18}{t}$; then $u \geq 2\sqrt{18} = 6\sqrt{2}$ and $t^2 + \dfrac{324}{t^2} = u^2 - 36$.

$OQ^2 = u^2 - 20u + 164 = (u-10)^2 + 64$. Since $10 \geq 6\sqrt{2}$, the value $u = 10$ is attained ($t^2 - 10t + 18 = 0$, $t = 5 \pm \sqrt{7}$), and $\min OQ^2 = 64$, $\min OQ = 8$.

Note that the nearest points are $\left(5 \pm \sqrt{7}, 5 \mp \sqrt{7}\right)$, not the vertex $(3\sqrt{2}, 3\sqrt{2})$ of the curve, where $OQ = \sqrt{2}(10 - 3\sqrt{2}) \approx 8.14$. (At the nearest points, OQ is normal to H: slope of OQ times slope of H is -1.)

$8 > 3$, so C and H do not meet. $\min PQ = 8 - 3 = 5$, and $\min PQ^2 = 25$.

Final Answer: 25

Question 33

English

For $t > 0$, let C_t be the circle with centre $(t^2, 0)$ and radius $\dfrac{27}{4t}$, and let $M(t)$ denote the greatest distance of the origin from the points of C_t. Then the least value of $M(t)$, $t > 0$, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$t > 0$ के लिए माना कि C_t केन्द्र $(t^2, 0)$ तथा त्रिज्या $\dfrac{27}{4t}$ वाला वृत्त है, तथा $M(t)$ मूलबिन्दु की C_t के बिन्दुओं से अधिकतम दूरी को निरूपित करता है। तब $t > 0$ के लिए $M(t)$ का न्यूनतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For a point at distance d from the centre of a circle of radius r, the greatest distance to the circle is $d + r$, attained at the end of the diameter through the point, whether the point is inside or outside.

Here $d = t^2$ and $r = \dfrac{27}{4t}$, so $M(t) = t^2 + \dfrac{27}{4t}$.

Split the second term into two equal parts so that the product of the three terms is constant: $M(t) = t^2 + \dfrac{27}{8t} + \dfrac{27}{8t}$.

By AM-GM, $M(t) \geq 3\sqrt[3]{t^2 \cdot \dfrac{27}{8t} \cdot \dfrac{27}{8t}} = 3\sqrt[3]{\dfrac{729}{64}} = 3 \cdot \dfrac{9}{4} = \dfrac{27}{4}$.

Equality holds when $t^2 = \dfrac{27}{8t}$, i.e. $t^3 = \dfrac{27}{8}$, $t = \dfrac{3}{2}$; then $M = \dfrac{9}{4} + \dfrac{9}{2} = \dfrac{27}{4}$.

(At $t = \dfrac{3}{2}$ the centre is $\left(\dfrac{9}{4}, 0\right)$ and the radius $\dfrac{9}{2}$, so the origin is inside C_t; the formula $d + r$ holds anyway.)

Least value $= 6.75$.

Final Answer: 6.75

Question 34

English

An ellipse has a focus at the origin, and the directrix corresponding to this focus is perpendicular to the x-axis. The ellipse passes through the points $(5, 0)$ and $(-15, 8)$. Then the eccentricity of the ellipse is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक दीर्घवृत्त की एक नाभि मूलबिन्दु पर है, तथा इस नाभि के संगत नियता x-अक्ष के लम्बवत है। दीर्घवृत्त बिन्दुओं $(5, 0)$ तथा $(-15, 8)$ से होकर जाता है। तब दीर्घवृत्त की उत्केन्द्रता _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let the directrix be $x = d$ ($d \neq 0$, since the focus is not on the directrix). For every point P of the ellipse, $PS = e \cdot PM$ with $S = (0, 0)$ and $PM = |x - d|$.

$(5, 0)$: $PS = 5$, so $5 = e|5 - d|$. $(-15, 8)$: $PS = \sqrt{225 + 64} = 17$, so $17 = e|15 + d|$.

Dividing: $5|15 + d| = 17|5 - d|$.

Case $5(15 + d) = -17(5 - d)$: $75 + 5d = -85 + 17d$, $d = \dfrac{40}{3}$; then $|5 - d| = \dfrac{25}{3}$ and $e = \dfrac{3}{5}$.

Case $5(15 + d) = 17(5 - d)$: $75 + 5d = 85 - 17d$, $d = \dfrac{5}{11}$; then $e = \dfrac{5}{|5 - 5/11|} = \dfrac{11}{10} > 1$. Here the two points are on opposite sides of the directrix, so this is a hyperbola, not an ellipse.

For the ellipse, $e = \dfrac{3}{5}$ with directrix $x = \dfrac{40}{3}$; both points and the focus lie on the same side of it, as they must for an ellipse.

Final Answer: 0.6

Question 35

English

A conic Σ has a focus at the origin O, the corresponding directrix is the line $3x + 4y = 75$, and Σ passes through the point $(3, 4)$. If (α, β) is the centre of Σ, then the value of $\alpha + \beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक शांकव Σ की एक नाभि मूलबिन्दु O पर है, संगत नियता रेखा $3x + 4y = 75$ है, तथा Σ बिन्दु $(3, 4)$ से होकर जाता है। यदि (α, β), Σ का केन्द्र है, तो $\alpha + \beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

By the focus-directrix definition, every point P of Σ satisfies $PS = e \cdot PM$, where PM is the distance from P to the directrix.

At $P = (3, 4)$: $PS = 5$ and $PM = \dfrac{|9 + 16 - 75|}{5} = 10$, so $e = \dfrac{1}{2} < 1$: Σ is an ellipse.

For an ellipse, the focus and the corresponding directrix are at distances ae and $\dfrac{a}{e}$ from the centre, on the same side, along the major axis. So the distance from focus to directrix is $\dfrac{a}{e} - ae = 2a - \dfrac{a}{2} = \dfrac{3a}{2}$.

Distance of O from $3x + 4y = 75$ is 15, so $\dfrac{3a}{2} = 15$, $a = 10$, $ae = 5$.

The major axis is the perpendicular from O to the directrix, direction $\dfrac{1}{5}(3, 4)$. The centre is at distance 5 from O on the side away from the directrix: $(\alpha, \beta) = -5 \cdot \dfrac{1}{5}(3, 4) = (-3, -4)$.

Check by expansion: $100(x^2+y^2) = (3x+4y-75)^2$ gives $91x^2 - 24xy + 84y^2 + 450x + 600y - 5625 = 0$; the partial derivatives $182x - 24y + 450$ and $-24x + 168y + 600$ vanish at $(-3, -4)$. (Also $(3, 4)$ is the vertex nearer the directrix: $a - ae = 5$.)

$\alpha + \beta = -7$.

Final Answer: -7